

MD. ARIFUL ISLAM

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Professional Summary

Software Engineer specializing in GPU-accelerated computing (CUDA C++) and applied Large Language Models (LLMs), with hands-on experience spanning real-time computer vision, high-performance computing, and Retrieval-Augmented Generation (RAG) systems. Proven record of delivering measurable performance gains (up to 50x speedup) in production imaging pipelines for semiconductor metrology and medical imaging, alongside research contributions in multimodal LLMs, vision-language models, and NLP. Currently pursuing further training in high-performance and AI systems.

Education

Chittagong University of Engineering and Technology (CUET)

B.Sc. in Computer Science and Engineering

Chittagong, Bangladesh

Jan 2018 – Jul 2023

- Final Grade: 3.21/4.00
- Thesis: TranSenA: A Transformer-based Framework for Sentiment Analysis of Restaurant Reviews

Work Experience

Acote Group

Software Engineer (Contractual)

Dhaka, Bangladesh

Jan 2026 – May 2026

- Leading the CUDA-based migration of a legacy DSP MRI image-reconstruction pipeline, re-engineering k-space-to-image conversion for parallel GPU execution to accelerate diagnostic imaging throughput.
- Architecting integration between an NVIDIA Jetson Xavier NX-based DPC Command Server and the MRI spectrometer controller, enabling low-latency k-space data processing and real-time image rendering.

Frontier Semiconductor Metrology (FSM) Bangladesh Ltd.

Software Engineer

Dhaka, Bangladesh

Jun 2023 – Present

- Engineered GPU-accelerated (CUDA C++) algorithms for wafer surface roughness computation, reducing processing time from 8 minutes to 9 seconds — a 50x performance improvement — using White Light Interference (WLI) technology with high-speed Basler machine-vision cameras.
- Built computer vision and OCR-based wafer ID recognition systems, enabling automated traceability across manufacturing pipelines.
- Developed customized stress-measurement solutions for film-deposited wafers for an international client (Sony, Japan) using 650nm and 780nm laser metrology.

SELISE Digital Platforms

Software Engineer Intern

Dhaka, Bangladesh

Oct 2022 – Nov 2022

- Designed and developed a full-stack e-commerce platform using Angular, Node.js, and MongoDB.

Projects

HW-Accelerated AutoFocus | NVIDIA Jetson Orin NX, CUDA C++, Basler Camera

Jan 2026 – May 2026

- Designed a hardware-accelerated autofocus system for real-time image processing and focus optimization; synchronized camera acquisition with motor-controller Z-axis traversal for automated focus sweeps.
- Built a client-server architecture to receive autofocus commands from FSM's RAFT software and coordinate camera-motor operations.
- Optimized autofocus computation using CUDA C++ kernels with edge-based sharpness metrics for low-latency focal-position determination.

Wafer Surface Topography | CUDA C++, HPC, Embedded Systems | GitHub

Sep 2024 – Present

- Integrated high-speed machine-vision cameras for real-time, high-volume image processing on embedded HPC systems.
- Parallelized image-processing routines using CUDA C++; developed batch processing to resolve memory-overflow issues with high-resolution imagery.

GPU-Accelerated MRI Image Reconstruction | CUDA C++, NVIDIA Jetson Xavier NX

Jan 2026 – May 2026

- Migrating a legacy DSP-based MRI reconstruction pipeline to CUDA-based parallel processing for k-space-to-image conversion.
- Integrating a Jetson Xavier NX DPC Command Server with the MRI spectrometer controller for real-time processing and image rendering.

- RAG-based LLM Chat Bot** | *LLM, RAG, Python* | Github

Jul 2025 – Present

 - Built a desktop application combining Retrieval-Augmented Generation (RAG) with a Large Language Model (LLM) for document-grounded question answering.
 - Added conversational memory to preserve context across multi-turn sessions.
 - Enabled PDF upload with answers synthesized from document content, session history, and the LLM’s pre-trained knowledge.
 - Ongoing optimization for faster, more secure response generation.

- Wafer ID Detection** | *Computer Vision, CNN, Pattern Recognition*

Feb 2024 – Aug 2024

 - Captured real-time wafer images using a high-speed Basler machine-vision camera.
 - Developed a CNN-based pattern recognition framework to detect semi-font wafer IDs.

Research Experience

- CUET NLP Lab**

Chittagong, Bangladesh

Research Assistant

Jul 2024 – Present

 - Optimizing RAG-based LLM systems for faster, more secure inference over multimodal embeddings and vision-language models.
 - Investigating cross-modal gender bias in hate-speech classification to improve fairness and transparency in vision-language models.

Publications

Md. Ariful Islam, Shawly Ahsan, Mohammed Moshikul Hoque. *TranSenA: A Transformer-based Framework for Sentiment Analysis of Restaurant Reviews*. IEEE International Conference on Signal Processing, Information, Communication & Systems (SPICSCON), 2024. iee

Technical Skills

GPU & HPC: CUDA C++, GPGPU programming, real-time parallel image processing, kernel optimization
AI / LLM: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), multimodal embeddings, vision-language models, conversational AI
ML / CV: Machine Learning, Deep Learning, Computer Vision, OpenCV, TensorFlow, Keras
Languages & Tools: C, C++, Python, Git, MongoDB, HTML/CSS, MFC
Data & Libraries: Pandas, Matplotlib, NumPy

Honors, Awards & Test Scores

International Collegiate Programming Contest (ICPC), ICPC Foundation — 2022
National Collegiate Programming Contest (NCPC), Bangladesh Computer Council — 2020
HSC General Scholarship Award, Board of Intermediate and Secondary Education, Dhaka — 2017
SSC General Scholarship Award, Board of Intermediate and Secondary Education, Cumilla — 2015
GRE: 309 (Quant: 161, Verbal: 148, AWA: 3.0) — Sep 2024
IELTS: 7.0 (L: 7.5, R: 7.5, S: 6.5, W: 6.0) — Nov 2024

Leadership & Volunteer Experience

Programming Coordinator, Computer Club, CUET (2021–2023) — Taught advanced data structures, number theory, and algorithms; organized contests and coordinated competitive programming teams.
Member, Greater Cumilla Student Welfare Association (2018–Present) — Served as student representative liaising with academia; assisted in medical fundraising for critically ill students.